SoundMatch-SR: A Benchmark for Synthetic–Real Sound-Effect Retrieval —Instance Correspondence Generalizes Across Categories, Category Structure Does Not

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Abstract

Perceptual studies report that listeners cannot reliably distinguish a well-made Foley or text-to-audio rendering of an event from a real recording of the same event [Nordahl et al., 2010]: the *identity* of the event survives the rendering. I ask whether modern audio embeddings represent that identity invariantly to how a sound was produced, and I find a sharp *dissociation*. I introduce **SoundMatch-SR**, a benchmark for cross-domain (synthetic–real) sound-effect retrieval, built on (i) the category-aligned DCASE-2023 Task-7 Foley corpus [Choi et al., 2023] (5,550 real and 25,900 synthetic clips from 37 generator systems, 7 classes) and (ii) a new 34-category benchmark labelled with the Universal Category System (FSD50K [Fonseca et al., 2022] real audio, CLAP-verified, with 10,420 instance-level Stable-Audio-Open [Evans et al., 2024b] twins). Off-the-shelf encoders carry a real, linearly-accessible synthetic–real gap (CLAP: 0.161 mAP; PANNs: 0.124; Proxy-A-distance 1.27). A lightweight head on frozen embeddings closes it *within* the training taxonomy ($0.161 \rightarrow 0.018$), but a *class-supervised* head does *not* transfer to unseen categories: held-out classes collapse below the frozen baseline, on both 7 and 34 categories. The remedy is to train on the *pairs*, not the labels: an instance-contrastive objective (a clip and its generated twin) learns to match a clip to its synthetic counterpart, and this *instance correspondence* generalizes to categories the head never saw—against the full test gallery (3,065 distractors, chance 0.03%) it retrieves the exact real twin with $\text{R@1} = 0.800$ (95% CI $[0.786, 0.813]$; vs. 0.611 frozen, 0.269 class-supervised), in all five folds and on three encoders (CLAP, PANNs, AST). Category *structure* transfers for no objective. Crucially, the effect requires *audio-conditioned* generation: text-only twins (ElevenLabs) carry no instance correspondence to recover, so the learned mapping is generator-specific, not a universal rendering inverse. I release the benchmark, the corpus recipe, the trained heads, and a reusable embedding transform.

1 Introduction

Foley research finds that well-synthesized everyday sounds are perceptually indistinguishable from recordings of the same action; the event's identity survives the synthesis-to-recording boundary for a human listener [Nordahl et al., 2010]. I make this measurable for machine representations:

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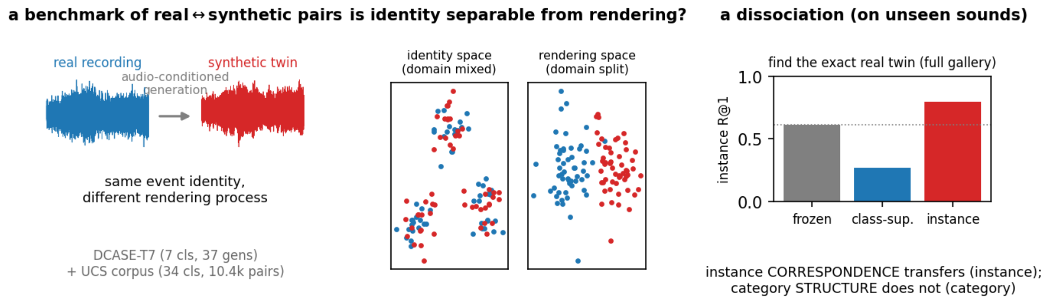

Figure 1: **SoundMatch-SR.** I pair each real recording with an audio-conditioned synthetic twin of the same event and ask whether an embedding can represent the event *identity* invariantly to the rendering. Synthetic–real *instance correspondence* transfers to sound types never seen in training (instance retrieval, right; full-gallery R@1), while category structure does not.

> *Is "sound identity" recoverable in embedding space, invariant to the rendering process—and if so, does that invariance transfer to sounds the model was not trained on?*

I operationalize it as retrieval rather than detection: detection rewards *exploiting* the synthetic–real gap, whereas retrieval and invariance reward *removing* it, which is the useful capability for sound-effect search, real/synthetic deduplication, and generative-audio evaluation. My answer is a dissociation that, to my knowledge, is new: the correspondence between a clip and its *audio-conditioned* synthetic twin is a learnable object that transfers across categories, while category identity is not—and this correspondence vanishes for text-only generators, so it is generator-specific rather than a universal rendering inverse.

Contributions. **(C1)** a two-part benchmark—a controlled 7-class corpus (DCASE-T7) and a diverse, instance-paired 34-category corpus—with leakage-safe splits and a reproducible generation recipe; **(C2)** a measured dissociation: an instance-contrastive objective generalizes synthetic–real *instance correspondence* to unseen categories (full-gallery R@1 0.800, all five folds), whereas class-supervised invariance does not (and degrades unseen classes below baseline); robust over five folds and three encoders, two of which did not participate in corpus verification; **(C3)** a fidelity-spectrum analysis showing the instance head recovers the mapping across twin-fidelity levels, collapsing only when the twin is independent of its source; **(C4)** released artifacts—benchmark, recipe, CLAP-verification, trained heads, and a reusable transform that adjusts any new clip's embedding.

2 Related Work

Audio-representation benchmarks. HEAR [Turian et al., 2022] and the recent massive embedding benchmarks MSEB [Heigold et al., 2026] and MAEB [El Assadi et al., 2026] evaluate frozen embeddings across classification, retrieval, and clustering, but treat all audio as one domain; none tests invariance to *how* a sound was produced. The encoders I benchmark—CLAP (Wu et al., 2023; Elizalde et al., 2023), PANNs [Kong et al., 2020], AST [Gong et al., 2021], and the SSL families BEATs [Chen et al., 2023], M2D [Niizumi et al., 2024], AudioMAE [Huang et al., 2022]—are evaluated frozen.

Synthetic-audio detection. Deepfake-environmental-audio detection is built on exactly the DCASE-T7 corpus [Ouajdi et al., 2024] and benchmarked in EnvSDD [Yin et al., 2025]; this pursues the mirror objective (exploit the gap). I remove it, and reproduce detection as a controlled *sensitive* head.

Text-to-audio generators. The synthetic domain comes from modern generators—Stable Audio Open [Evans et al., 2024b] and its predecessor [Evans et al., 2024a], AudioGen [Kreuk et al., 2023], AudioLDM [Liu et al., 2023]—whose realism is typically scored by Fréchet Audio Distance [Kilgour et al., 2019]; my sensitive head offers a complementary, paired realness axis.

Domain adaptation and contrastive learning. I draw on Proxy-A-distance [Ben-David et al., 2010], domain-adversarial training [Ganin and Lempitsky, 2015], Deep CORAL [Sun and Saenko, 2016], supervised contrastive learning [Khosla et al., 2020], and IRM [Arjovsky et al., 2019]. My generalizing objective is instance-level in the sense of instance discrimination [Wu et al., 2018] and contrastive representation learning (Chen et al., 2020b; Radford et al., 2021); the negative result—that class-level supervision does not transfer the invariance—is a domain-generalization phenomenon.

3 The SoundMatch-SR Benchmark

Core corpus (DCASE-T7). The DCASE-2023 Task-7 corpus [Choi et al., 2023] is a ready-made, category-aligned real/synthetic set across 7 classes: 5,550 real clips (sourced from FSD50K, UrbanSound8K [Salamon et al., 2014], BBC, and Freesound; I use DCASE's source-disjoint dev/eval as train-val/test) and 25,900 synthetic clips from 37 generator systems. It is the controlled setting for a clean gap measurement and a leave-one-class-out probe.

UCS corpus. To test generalization I need many categories and instance-level pairs. I label real audio with the Universal Category System (UCS), mapping the 200 AudioSet [Gemmeke et al., 2017] labels of FSD50K onto 34 UCS categories spanning all morphologies (impact, texture, tonal, whoosh, vocal, mechanical, ambience). I *verify* every clip with zero-shot CLAP [Wu et al., 2023] (keep iff its label is top-5 by audio–text cosine), retaining 10,420/13,579 (77%) and removing FSD50K's noisiest multi-labels. For each verified anchor I generate a Stable-Audio-Open *audio-init* twin [Evans et al., 2024b] conditioned on the real clip, yielding 10,420 instance pairs. Splits follow FSD50K's train/val/eval, keyed on the Freesound uploader so crops of one recording never straddle the boundary.

Tasks and metrics. *Category retrieval*: query a clip; rank opposite-domain clips; relevant iff same category. I report mAP and a same-domain control, so the headline *domain gap* = control − cross. *Instance retrieval* (UCS): relevant iff the exact cross-domain twin (shared instance id); R@1 and MRR. I report bootstrap CIs, and on the UCS corpus a k-fold leave-classes-out protocol (Sec. 5). Full per-class, per-split statistics and the datasheet are in the appendix.

4 Method

I train a small MLP head (CLAP-512 $\rightarrow 256$, ℓ_2-normalised) on *frozen* embeddings; the head is the adjusted embedding and applies to any new clip. The objective decides what it learns. The **invariant** head is class-supervised contrastive with cross-domain positives up-weighted (optionally with DANN/IRM): it pulls same-category clips together and removes domain. The **sensitive** head is domain-supervised within category: the deliberate mirror, organising the space by rendering—its real-vs-synthetic direction is a fidelity axis. The **instance** head uses as positive a clip and *its own* generated twin (with pair-aware batching so twins co-occur in a batch); it learns the synthetic-to-real *mapping* rather than class identity.

5 Experiments

Frozen encoders carry a real gap (Table 2). The synthetic–real axis is real and linearly accessible in both CLAP and PANNs [Kong et al., 2020].

Within-taxonomy the gap closes; the two heads are mirror images. The invariant head closes the DCASE gap $0.161 \rightarrow 0.018$ while *raising* same-domain control ($0.935 \rightarrow 0.988$) and making the space event-organised (silhouette by event $0.18 \rightarrow 0.75$, by domain $0.04 \rightarrow 0.00$); the sensitive head flips it to organise by rendering (PAD 1.88, silhouette by domain 0.75). Cross-domain supervised contrast alone closes the gap; DANN/CORAL/IRM add < 0.002 (Table 1); the geometry flip is visible in Fig. 2.

The closed-world failure. Holding one class out of training and evaluating on it, the held-out class collapses *below* frozen (keyboard $0.69 \rightarrow 0.43$, gunshot $0.81 \rightarrow 0.31$, rain $0.75 \rightarrow 0.30$). Adding categories does not fix it (34-class: $0.40 \rightarrow 0.19$).

Table 1: The two heads on DCASE-T7 (left) and the bridging ablation (right). The invariant head closes the gap and organises the space by event; the sensitive head is its mirror. Cross-domain supervised contrast does the work.

head	control	cross mAP	sil(ev/dom)	ablation	gap	PAD
frozen	0.935	0.774	0.18 / 0.04	supcon only	+0.019	1.38
invariant	0.988	0.970	0.75 / 0.00	+CORAL	+0.017	1.38
sensitive	0.206	0.157	−0.04 / 0.75	+DANN	+0.019	1.38
				+DANN+IRM	+0.018	1.38

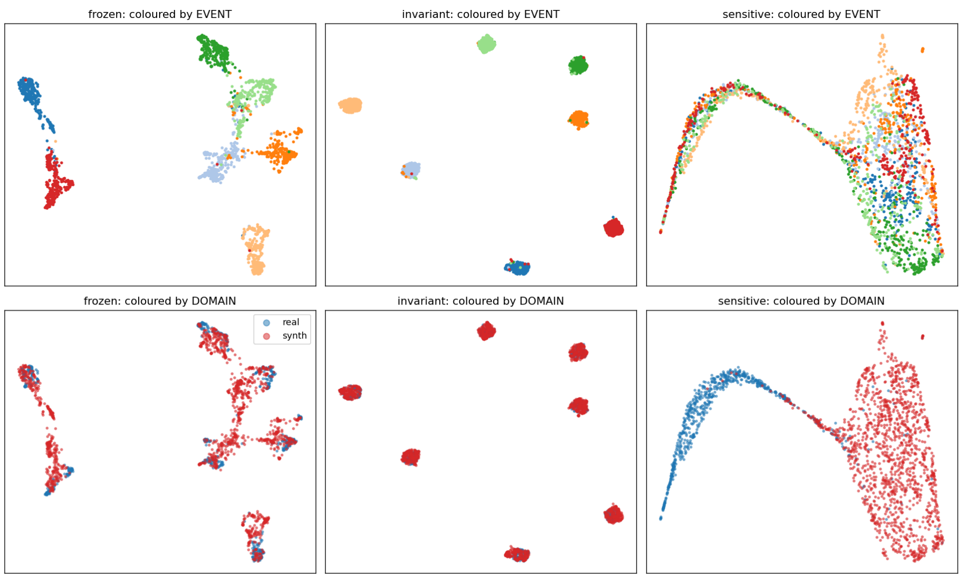

Figure 2: UMAP of the DCASE-T7 test set, frozen vs. the two heads, coloured by event (top) and by domain (bottom). The invariant head tightens event clusters with real/synth overlapping within each; the sensitive head splits real from synth and smears events—the geometry flips.

The dissociation (Table 3, Fig. 3a). Training on the pairs, the instance objective generalizes instance correspondence to categories unseen *by the head* (the frozen backbone's pretraining is uncontrolled, so "unseen" means not seen during head training). Against the full real test gallery ($N = 3{,}065$ distractors of *all* categories, chance R@1 $= 0.0003$) the instance head retrieves the exact real twin with R@1 0.800 ($[0.786, 0.813]$), beating frozen (0.611) in all five folds (min margin $+0.148$) and class-supervision (0.269, which *hurts*) in all five (min $+0.493$); no objective transfers category clustering (all $\leq$ frozen on unseen-category mAP). A compute-matched supervised cross-entropy classifier head fares identically to class-supcon (0.282), so the failure is from training on *class labels*, not from the contrastive form—only instance-pair supervision generalizes. It holds on three encoders—CLAP, PANNs, and AST [Gong et al., 2021] (full-gallery instance R@1 $0.80/0.69/0.93$ vs. frozen $0.58/0.62/0.82$ vs. class $0.25/0.32/0.29$; instance $>$ frozen and class $\ll$ frozen on all three). Two of these (PANNs, AST) did *not* participate in CLAP-verification of the corpus, so the dissociation is not an artifact of the CLAP filter. (AST's frozen R@1 is already high—these audio-init twins are near-duplicates for a strong encoder—which is exactly why the class-supervision *collapse* to 0.29 is the more informative half of the dissociation.)

Not near-copies (Fig. 3b). Audio-init twins resemble their source, so I test whether the head merely exploits near-copies: across twin-fidelity levels (the head still helps, advantage $+0.43..0.52$ over frozen) it holds up and collapses to chance only when the twin is independent of its source. (This control is measured in CLAP space and is in-domain; the across-category claim rests on the leave-classes-out result above.)

Table 2: Frozen encoders: synthetic–real gap on DCASE-T7 (test split).

frozen	control	synth→real mAP	gap	PAD	domain-probe
CLAP	0.935	0.774 [0.766,0.783]	+0.161	1.27	0.90
PANNs CNN14	0.798	0.674	+0.124	1.39	0.90

Table 3: Five-fold leave-classes-out on the UCS corpus, CLAP, held-out (unseen) categories. Instance-R@1 is against the *full* real test gallery ($N = 3{,}065$, chance 0.0003); 95% bootstrap CIs over queries. category-mAP is full-gallery synth→real category retrieval.

variant (unseen categories)	category-mAP	**instance-R@1 (full gallery)**
frozen	0.611	0.611 $[0.594, 0.629]$
class-supcon	0.459	0.269 $[0.254, 0.285]$
class-CE (classifier)	0.451	0.282 $[0.266, 0.297]$
instance	0.492	$\mathbf{0.800}$ $[\mathbf{0.786}, \mathbf{0.813}]$

Cross-generator scope (the claim's boundary). Adding 670 ElevenLabs (text-only) twins captioned from each clip's metadata, instance correspondence is weak (frozen R@1 0.11 vs. 0.55–0.98 for audio-init): the caption is a bottleneck, so the twins are category-appropriate but not renderings of the *specific* clip. The learned mapping is therefore specific to *audio-conditioned* generation—it is an instance-correspondence inverse for a given generator family, not a universal synthetic→real transform.

6 Conclusions, Limitations, and Future Work

Audio-conditioned synthetic→real instance correspondence is learnable and transfers across categories; category identity is not, and the correspondence is generator-specific. **The central caveat** (stated up front): because the synthetic twins are generated *from* the source audio, "retrieve the exact twin" is closer to cross-domain near-duplicate matching than to inverting a general rendering process—the ElevenLabs result (text-only twins, R@1 0.11) shows the learned mapping does not transfer to a generator that does not condition on the source. **Other limitations**: category clustering does not transfer to unseen categories (train per target taxonomy); twins are Stable-Audio audio-init at one operating point per corpus; frozen-embedding heads cannot recover information the backbone discarded; instance pairing requires audio-conditioned generation; and the UCS corpus is CLAP-verified, which could bias CLAP results upward—I mitigate this by showing the dissociation holds on PANNs and AST (which did not verify the corpus), and by noting that the DCASE-T7 gap (Table 2, an *unverified* corpus) is comparable. SSL encoders (BEATs [Chen et al., 2023], AudioMAE [Huang et al., 2022]) are provided as activatable loaders. **Future work**: broader real sources (e.g. VGGSound [Chen et al., 2020a]) and domain-relevant corpora (game and film SFX), and using the sensitive head as a realness metric for generative-audio evaluation.

Reproducibility and release

Code (MIT), DCASE and UCS manifests, the CLAP-verification, the generation recipe (event, prompt, seed, steps—bit-reproducible without redistributing restricted audio), the trained heads, and a reusable transform are released. A datasheet [Gebru et al., 2021] and a persistent hosting/maintenance plan are in the appendix.

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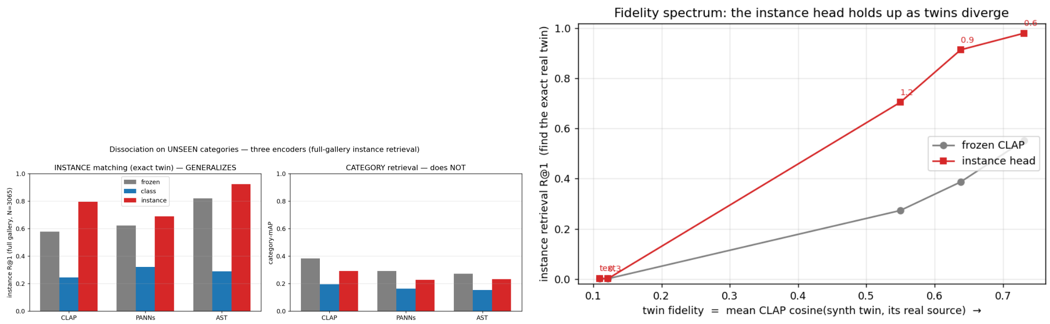

(a) (b)

Figure 3: (a) The instance/category dissociation on unseen categories, three encoders: instance matching generalizes (left), category retrieval does not (right). (b) Fidelity spectrum: the instance head holds up as twins diverge from their source.

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A Datasheet, hosting, and maintenance

The appendix provides the Gebru et al. [Gebru et al., 2021] datasheet (motivation, composition, collection, preprocessing, uses, distribution), per-class/per-split statistics, the licensing matrix (CC-licensed audio redistributed; BBC and restricted audio shipped as a fetch manifest plus the generation recipe), and the maintenance plan (versioned manifests; the code regenerates all derived artifacts from the manifest and the source archives).

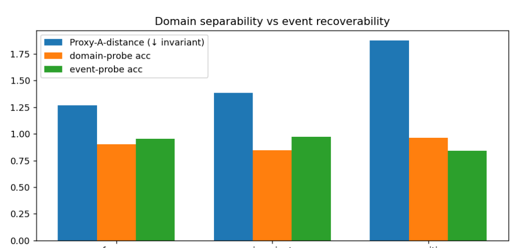
B Corpus statistics and per-event analysis

Table 4: Per-domain, per-split clip counts. DCASE-T7 real test is DCASE's source-disjoint eval set; the UCS corpus uses FSD50K's train/val/eval, uploader-keyed.

corpus	domain	train	val	test
DCASE-T7	real	4,150	700	700
DCASE-T7	synth	18,165	3,884	3,851
UCS (34 cls)	real	8,315	999	4,265
UCS (34 cls)	synth (twins)	6,636	719	3,065

Table 5: Per-event frozen-CLAP cross-domain (synth→real) mAP on DCASE-T7. Fast, granular transient/mechanical sounds transfer worst.

	motor	gunshot	dog_bark	sneeze	rain	footstep	keyboard
mAP	0.852	0.810	0.809	0.793	0.752	0.718	0.692



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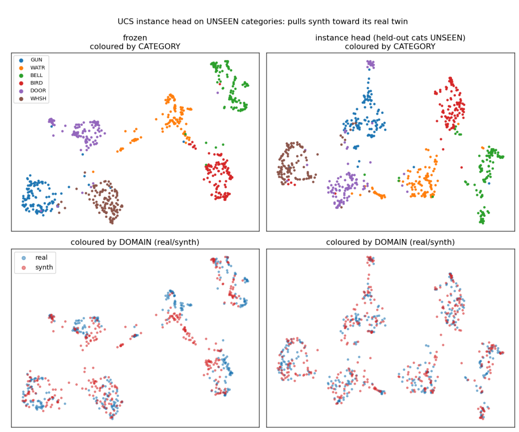

(b)

Figure 4: (a) Proxy-A-distance and event/domain linear-probe accuracy across the DCASE heads. (b) UMAP of the instance head on unseen UCS categories: it mixes real/synth (domain) more than frozen.